

The solver does not deliver the β_{sub0} it is given

Diagnosis and remediation plan for `lunar_regolith_DSM`

Working note — evidence from batch 2026-08-07__07.26.38_wedge_bc

2026-08-25

Summary — confirmed by experiment, 2026-08-26. The realized kinetic coefficient is $1.22\times$ the requested `-beta_sub0`, so every run sublimates and deposits 18% slower than intended. The capillary length d_0 is exact. The excess is precisely the two Karma–Plapp thin-interface corrections that `lunar_main.c:829` adds to τ_{sub} for the asymptotics to subtract back off — and which the measurement says are never subtracted. One of the two (the thermal one, 13.1 of the 17.8 points) is wrong on its face and should be fixed regardless. The fix is to τ_{sub} , not to the Gibbs–Thomson prediction, which is correct as written.

A four-arm β_{sub0} ladder (§6) has since confirmed this over a $40\times$ range: the measured ratio moves $0.32 \rightarrow 0.95$ exactly as predicted, and $\beta_{\text{fit}}/\beta_{\text{bare}} = 1.0009 \pm 0.0007$ throughout.

1. What was measured

The wedge geometry places an apex-centered ice annulus in a tapering pore with the virtual apex at $y = L_y/2$, so on the centerline the interface normal is *exactly* $\pm\hat{x}$ and v_n is a one-dimensional displacement rate. `postprocess/wedge_gt_velocity.py` measures it on both menisci of all nine runs and compares against

$$v_n = \frac{\sigma \rho_{vs}(T)/\rho_{vs}(T_0) - d_{0,\text{sub0}} \chi}{\beta_{\text{sub0}}}, \quad \sigma = \frac{\rho_v - \rho_{vs}(T)}{\rho_{vs}(T)}. \quad (1)$$

The solver recovers 0.826 (inner) / 0.811 (outer) of that rate — flat over 90 days, and flat across nine vapor-boundary combinations whose residual driving force $\sigma - d_0\chi$ spans two orders of magnitude. Interface-extraction noise is 0.5% against an 18% effect, and three independent v_n estimators agree on the median to 1%, so the shortfall is not a measurement artifact.

2. The regression: which coefficient is wrong

Rather than assume Eq. (1) and read off a ratio, fit both coefficients freely. Every snapshot supplies a triple (σ, v_n, χ) , so

$$\sigma = \beta v_n + d_0 \chi \quad (\text{no intercept}) \quad (2)$$

is an ordinary two-column least-squares problem. **The design is identifiable because the two menisci carry opposite-sign curvature** — the inner one is concave, the outer convex — while the 3×3 reservoir grid moves v_n at nearly fixed χ . Two genuinely different directions:

χ spans	$-5.259 \times 10^3 \dots +3.459 \times 10^3 \text{ m}^{-1}$
v_n spans	$-1.459 \times 10^{-12} \dots +1.378 \times 10^{-12} \text{ m/s}$
$\text{corr}(v_n, \chi)$	-0.7095 (correlated, but far from collinear)
condition number	2.58

Fitting $n = 3492$ points (nine runs $\times$ two fronts $\times$ every fifth snapshot after the initial transient):

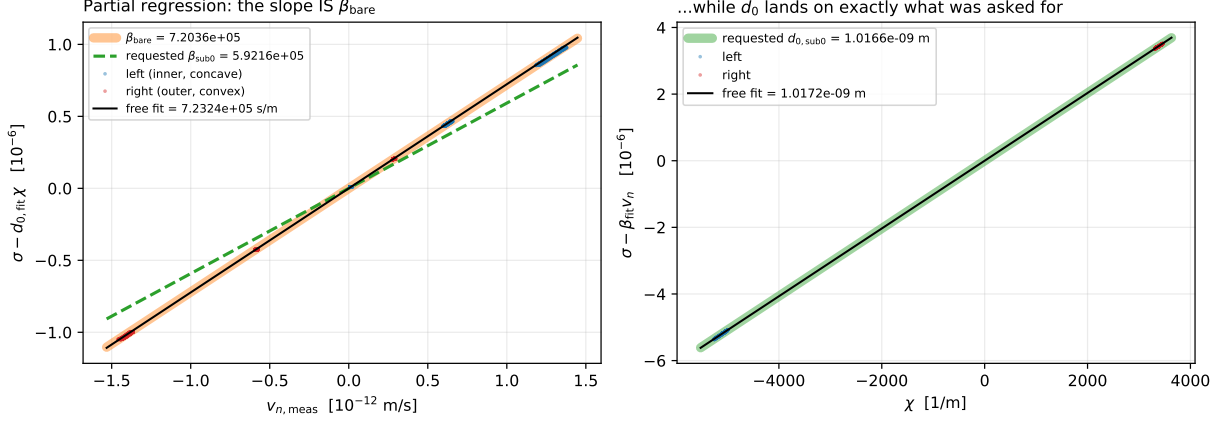


Figure 1: Partial-regression views of the same fit. **Left:** with the fitted curvature term removed, $\sigma - d_0\chi$ against v_n is a straight line through the origin whose slope is the realized β . The black fit lies inside the β_{bare} band (orange) and is visibly steeper than the requested β_{sub0} (green dashed). **Right:** with the kinetic term removed, the slope against χ falls inside the $d_{0,\text{sub0}}$ band: the capillary length is delivered exactly. Both fronts, all nine runs, every fifth snapshot.

	β [s/m]	$\times \beta_{\text{sub0}}$	d_0 [m]	$\times d_{0,\text{sub0}}$
pooled, both fronts	$7.2324 \times 10^5 \pm 1.8 \times 10^2$	1.2214	$1.0172 \times 10^{-9} \pm 3.5 \times 10^{-14}$	1.0006
left front only	$7.1384 \times 10^5 \pm 1.6 \times 10^2$	1.2055	$1.0166 \times 10^{-9} \pm 2.7 \times 10^{-14}$	1.0000
right front only	$7.2716 \times 10^5 \pm 1.3 \times 10^2$	1.2280	$1.0167 \times 10^{-9} \pm 3.5 \times 10^{-14}$	1.0001

$R^2 = 0.999997$ pooled; the rms residual is 5.4×10^{-9} , i.e. 0.14% of $|\sigma|$. Holding $d_{0,\text{sub0}}$ fixed and fitting β alone gives 7.2106×10^5 , within 0.3% of the free fit — the two parameters are not trading off against each other.

Robustness. Leave-one-run-out (drop each of the nine in turn and refit) moves β over 7.2204 – 7.2419×10^5 , a spread of 0.30% of the mean. No single vapor-BC condition drives the result. The residual left/right split (1.206 vs 1.228) is $\pm 1\%$ and is consistent with the quoted σ -extraction sensitivity (0.7% left, 0.3% right).

Conclusion. $d_{0,\text{sub0}}$ is recovered to 0.06%. The Gibbs–Thomson *curvature* physics is exact and the prediction formula is right. The entire deficit lives in β , and it is a clean multiplicative factor of 1.22.

3. β lands on the *bare* kinetic coefficient

Dividing the phase residual by $3 \text{ mob}/\epsilon$ gives the Moure & Fu form

$$\tau_{\text{sub}} \partial_t \phi = \epsilon^2 \nabla^2 \phi - f'(\phi) + \lambda_{\text{sub}} g(\phi) \frac{\rho_{vs}(T)}{\rho_{\text{ice}}} \sigma, \quad (3)$$

whose sharp-interface limit is exactly the Gibbs–Thomson condition Eq. (1), with the two coefficients it realizes given by

$$d_{0,\text{realized}} = d_{0,\text{sub0}}, \quad \beta_{\text{realized}} = \underbrace{\frac{\tau_{\text{sub}} d_{0,\text{sub0}}}{\epsilon^2}}_{\beta_{\text{bare}}} - C_{\text{corr}}. \quad (4)$$

C_{corr} is an $O(\epsilon)$ term generated by σ varying *inside* the diffuse interface. `lunar_main.c:829` pre-inflates τ_{sub} by exactly what it assumes C_{corr} to be, so that the two cancel and $\beta_{\text{realized}} = \beta_{\text{sub0}}$:

$$\tau_{\text{sub}} = \epsilon \lambda_{\text{sub}} \left(\underbrace{\frac{\beta_{\text{sub}}}{a_1}}_{82.2\%} + \underbrace{\frac{a_2 \epsilon}{\text{diff_sub}}}_{13.1\%} + \underbrace{\frac{a_2 \epsilon}{\text{dif_vap}}}_{4.7\%} \right), \quad \beta_{\text{bare}} = 7.2036 \times 10^5 \text{ s/m}. \quad (5)$$

(The bracket is written in the $\rho_{vs}/\rho_{\text{ice}}$ -scaled variables the code carries internally; β_{bare} above is already converted back to the unscaled units of β_{sub0} .) The fit gives $\beta/\beta_{\text{bare}} = 1.0040$. $C_{\text{corr}} \approx 0$: **nothing is subtracted, so the inflation survives intact**, and $\beta_{\text{bare}}/\beta_{\text{sub0}} = 1.2165 \Rightarrow$ ratio 0.8220 against a measured 0.826/0.811.

This is a defect in a *model parameter definition*. Eq. (1) is the target physics and is correct; the solver simply does not realize the β it was handed.

4. Δ : what it is, and why it matters

Definition. Δ is the amount by which the realized kinetic coefficient exceeds the requested one. From Eq. (5) with $C_{\text{corr}} = 0$:

$$\Delta \equiv \beta_{\text{bare}} - \beta_{\text{sub0}} = a_1 a_2 \epsilon \left(\frac{1}{\text{diff_sub}} + \frac{1}{\text{dif_vap}} \right) \frac{\rho_{\text{ice}}}{\rho_{vs}(T)} \quad (6)$$

in the unscaled units of β_{sub0} (s/m). The bracket is the pair of a_2 terms τ_{sub} carries; a_1 converts the bracket to a kinetic coefficient; the $\rho_{\text{ice}}/\rho_{vs}$ factor undoes the scaling the code works in internally. At the wedge parameters ($\epsilon = 8.584 \times 10^{-7}$ m, $T = -20^\circ\text{C}$):

	value [s/m]	share of Δ
thermal, $a_1 a_2 \epsilon \rho_{\text{ice}}/(\rho_{vs} \text{diff_sub})$	9.44×10^4	73.7 %
vapor, $a_1 a_2 \epsilon \rho_{\text{ice}}/(\rho_{vs} \text{dif_vap})$	3.37×10^4	26.3 %
Δ	1.282×10^5	
β_{sub0} (requested)	5.9216×10^5	
$\beta_{\text{bare}} = \beta_{\text{sub0}} + \Delta$ (realized)	7.2036×10^5	

Why it matters — three properties, none of them benign.

(i) *It is additive, not multiplicative.* The shortfall is $\beta_{\text{sub0}}/(\beta_{\text{sub0}} + \Delta)$, so it is not a fixed factor: it depends on how large β_{sub0} itself is. The ladder measured exactly this, $0.32 \rightarrow 0.95$ across four arms (§6). Any “just scale β by the observed ratio” fix is therefore calibrated to one parameter point and wrong everywhere else — see §7.

(ii) *It scales as ϵ and as $1/\rho_{vs}(T)$.* Coarser interfaces and colder temperatures both make it worse, and neither is a free knob: ϵ is set by resolution requirements and T by the science. At Molaro’s $\epsilon = 1.2 \times 10^{-7}$ m with $\alpha_c = 0.1$, $\Delta/\beta_{\text{sub0}} = 0.23$; at the wedge’s it is 0.22; at -60°C it is 20.7.

(iii) *It is a hard floor on the kinetics the model can represent.* Since $\beta_{\text{realized}} = \beta_{\text{sub0}} + \Delta \geq \Delta$ for any positive β_{sub0} , and $\beta \propto 1/\alpha_c$, there is a **ceiling on the attachment coefficient** no input can exceed:

T [$^\circ\text{C}$]	ρ_{vs} [kg/m^3]	Δ [s/m]	largest reachable α_c
−60	8.885×10^{-6}	1.225×10^7	0.0006
−40	1.055×10^{-4}	1.031×10^6	0.0077
−20	8.487×10^{-4}	1.282×10^5	0.0619
0	5.047×10^{-3}	2.156×10^4	0.3684

(at $\epsilon = 8.584 \times 10^{-7}$ m; halving ϵ halves every Δ and doubles every ceiling.) In the Molaro configuration this bites directly: the requested $\alpha_c = 0.1$ is delivered as 0.081, and at $\alpha_c \rightarrow 1$ — the end one would want for an attachment-limited Demmenie replication — Δ *exceeds* $\beta_{\text{sub}0}$ by $2.3\times$.

5. Why $C_{\text{corr}} = 0$: the inner-region analysis

5.1 What the correction actually is

It is not physics. At finite ϵ the diffuse model produces a *spurious* addition to the sharp-interface kinetic coefficient, because σ varies inside the interface where the phase-change source lives. τ_{sub} is inflated by exactly that spurious amount so the two cancel and the model realizes $\beta_{\text{sub}0}$. The construction is right; the question is only whether this model produces the spurious term it is being compensated for.

5.2 The solvability condition

Stretch the interface: ζ is signed normal distance (positive into the vapor), $\eta = \zeta/\epsilon$. At leading order the phase equation gives the equilibrium profile $\phi_0 = \frac{1}{2}(1 + \tanh(\eta/2))$. At the next order the linear operator is singular, and the Fredholm condition — projecting onto the translation mode $\partial_\eta \phi_0$ — is what produces the interface condition. Writing $\sigma = \sigma_{\text{int}} + \delta\sigma(\eta)$ with σ_{int} the outer solution extrapolated to $\zeta = 0$, the two integrals that set the coefficients are

$$\int (\partial_\eta \phi_0)^2 d\eta = \frac{1}{6}, \quad \int g(\phi_0) \partial_\eta \phi_0 d\eta = \frac{1}{30}, \quad a_1 = \frac{1/6}{1/30} = \mathbf{5} \quad (7)$$

— verified numerically to six digits, and exactly the a_1 the code uses. The correction rides on a *third* integral,

$$I = \int g(\phi_0) \partial_\eta \phi_0 \delta\sigma(\eta) d\eta, \quad (8)$$

and the entire question is whether I is zero.

The weight $g(\phi_0) \partial_\eta \phi_0$ is even in η , because $\phi_0(-\eta) = 1 - \phi_0(\eta)$ makes $g = \phi_0^2(1 - \phi_0)^2$ even and $\partial_\eta \phi_0 = \frac{1}{4}\text{sech}^2(\eta/2)$ even. So I picks out the *even* part of $\delta\sigma$ and is blind to the odd part.

5.3 Symmetric D : the deviation is even, and $I \neq 0$

With D equal on both sides and the flux splitting $\pm j/2$, the quasi-static inner balance $\partial_{\eta\eta}^2 \rho_v \propto \partial_\eta \phi_0$ integrates to $\partial_\eta \rho_v \propto (\phi_0 - \frac{1}{2})$, which is *odd*; so $\delta\sigma \propto \Psi(\eta) = \ln \cosh(\eta/2) - |\eta|/2 + \ln 2$ is *even* and decays to zero at both infinities. Even deviation against even weight: $I \neq 0$, and this is where Karma & Rappel's $-a_1 a_2 \epsilon / D$ comes from.

5.4 One-sided $D_v \phi_a$: the deviation is identically zero

Our vapor equation carries $D_v \phi_a$, which vanishes in the ice. Dropping storage (justified: $v_n \epsilon / (\xi_v D_v) = 2.4 \times 10^{-11}$) the inner balance is

$$\partial_\eta (D_v \phi_a \partial_\eta \rho_v) = -\epsilon v_n \rho_{\text{ice}} \partial_\eta \phi_0. \quad (9)$$

Integrate once; no flux into the ice fixes the constant at $C = \epsilon v_n \rho_{\text{ice}}$:

$$D_v \phi_a \partial_\eta \rho_v = \epsilon v_n \rho_{\text{ice}} (1 - \phi_0) = \epsilon v_n \rho_{\text{ice}} \phi_a \implies \partial_\eta \rho_v = \frac{\epsilon v_n \rho_{\text{ice}}}{D_v} \quad \text{wherever } \phi_a \neq 0. \quad (10)$$

The ϕ_a cancels. And that constant is precisely the outer Stefan gradient — $D_v \partial_\zeta \rho_v|_{\text{int}} = v_n \rho_{\text{ice}}$ gives the same value. The inner solution *is* the outer solution continued through the interface, so

$$\boxed{\delta\sigma \equiv 0 \implies I = 0 \implies C_{\text{corr}} = 0} \quad (11)$$

This is a *vanishing theorem*, not a cancellation: $\delta\sigma$ is not merely odd, it is null, so $I = 0$ for *any* weight and no prefactor is needed. (Deep in the ice the `air_lim` = 10^{-6} floor on ϕ_a breaks the cancellation, but $g(\phi_0)$ there is $\lesssim 10^{-12}$ and contributes nothing.)

5.5 An honest limit on this derivation

The vanishing result above is robust because it needs no prefactor. The *symmetric* branch is not equally settled here: constructing $\delta\sigma = \Psi$ as above and evaluating Eq. (8) gives $a_2 = -6I = 0.087$, against Karma & Rappel’s 0.158. So this sketch is missing $O(\epsilon)$ contributions — the curvature term, the ϕ_1 back-reaction, or the precise matching of the flux split. **We have not re-derived a_2 , and should not claim to.** What is established is the structural contrast: even-and-nonzero versus identically-null. The measurement supplies the rest.

5.6 The thermal channel fails for a different reason

The thermal term is not a one-sidedness question — k is interpolated and nonzero in both phases, so a correction does exist there. It is wrong because `a2*eps/diff_sub` assumes latent heat drives σ at *unit strength*. Temperature reaches σ only through $\rho_{vs}(T)$, so the channel carries a Clausius–Clapeyron factor. Reading off $d\sigma/d(v_n\epsilon)$ per channel:

channel	correct coefficient	value
vapor	$1/D_v$ <i>coded:</i> <code>1/dif_vap</code>	5.27×10^4 4.59×10^4 ✓ form, but $I = 0$
thermal	$(d\rho_{vs}/dT) L_{\text{sub}}/k$ <i>coded:</i> <code>1/diff_sub</code>	1.0×10^2 (k_{ice}) – 1.2×10^4 (k_{air}) 1.29×10^5 × over-weighted 11–1300×

`ThermalCond` interpolates between the two k ’s, so *any* reasonable choice leaves this term negligible. This defect stands independently of everything above.

5.7 What the measurement bounds — and what it does not

spurious term	form	seen by the ladder?
kinetic correction	$\propto v_n$	yes — $\leq 0.1\%$ of β_{bare}
solute trapping	$\propto \epsilon v_n$ (shifts σ_{int})	yes — also $\propto v_n$, absorbed into fitted β
surface diffusion	tangential flux along the interface	no — different operator

The first two are dead. The third is invisible to a centerline v_n measurement and is *not* bounded by anything in this note. One reassurance and one caveat: the surface excess of the diffusivity weight, $\int [(1 - \phi_0) - \Theta(-\eta)] d\eta$, is *exactly zero* by the same antisymmetry, which suggests the spurious surface diffusion is suppressed too — but that is a suggestion, not the calculation. It matters because the model has *no* surface-diffusion term by construction, so a spurious one would appear nowhere in the source, and the Molaro campaign is separately reasoning about *physical* surface diffusion. The two must not be conflated.

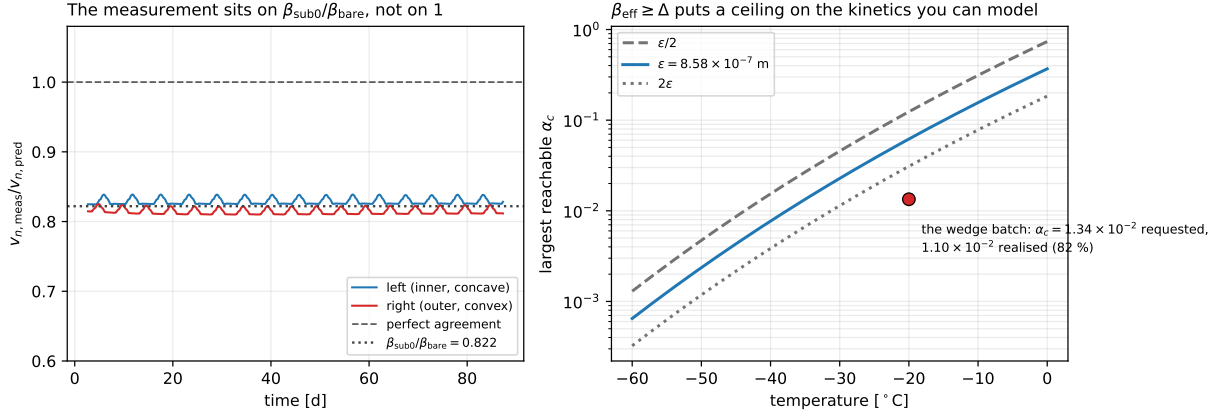


Figure 2: **Left:** the measured ratio tracks $\beta_{\text{sub}0}/\beta_{\text{bare}}$ for the whole run on both fronts. **Right:** the ceiling on the condensation coefficient implied by $\beta_{\text{eff}} \geq \Delta$, as a function of temperature and ϵ .

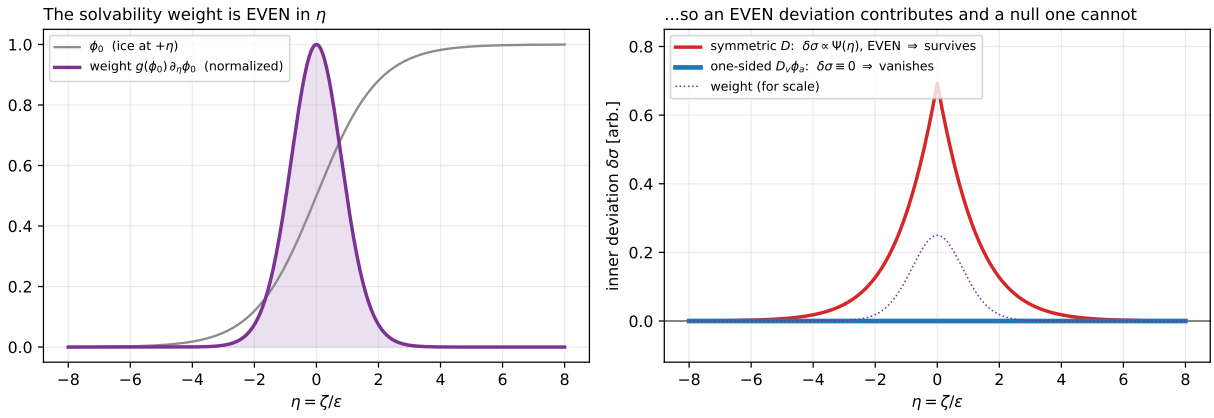


Figure 3: The structure behind §5. **Left:** the solvability weight $g(\phi_0)\partial_\eta\phi_0$ is even in η , so it sees only the even part of the inner deviation. **Right:** with symmetric D the deviation is even and non-zero, and the correction survives; with the one-sided $D_v\phi_a$ it is identically zero, so no correction can arise for any weight.

5.8 What has been ruled out

- **Grid pinning / lattice friction.** The deficit is a constant *factor*, identical on fronts whose speeds differ $60\times$. A threshold or friction cannot behave that way.
- **Interface-extraction bias.** Three independent v_n estimators agree on the median to 1%; the residual scatter is 0.5%, zero-mean, and locked to sub-cell phase.
- **A mis-converted a_1 or a_2 .** ϵ here is Moure & Fu's ϵ , and is deliberately *not* Kaempfer & Plapp's W ($W_{\text{KP}} = \sqrt{2}\epsilon$); the constants already absorb the difference. Mapping to K&P's $[-1, 1]$ form gives $d_{0, \text{sub}} = a_1^{\text{KP}}\sqrt{2} \cdot 4 \cdot \epsilon/\lambda_{\text{sub}} = 5.0000\epsilon/\lambda_{\text{sub}}$ against the coded $a_1 = 5.0$, and $a_1^{\text{KP}}a_2^{\text{KP}}\sqrt{2} = 0.78338$ against the coded $a_1a_2 = 0.7905$ (0.91%). This is already documented at `enceladus_DSM/src/assembly.c:46-58`. The free fit recovering $d_{0, \text{sub}0}$ to 0.06% is an independent measurement of a_1 .
- ξ_v . It multiplies the vapor diffusion and the ice source identically, so it cancels between them in the inner balance (§5) and survives only in storage/diffusion, $v_n\epsilon/(\xi_v D_v) = 2.4 \times 10^{-11}$. The unscaled `dif_vap` in Eq. (5) is the correct choice; substituting $\xi_v\text{dif_vap}$ would put the bracket at 6.36 instead of 0.133 and predict a $58\times$ shortfall.

6. The ladder confirms it

The diagnosis above rested on one $(\epsilon, \beta_{\text{sub0}}, T)$ point. Because the inflation is additive and β_{sub0} is not, the predicted shortfall is a strong function of β_{sub0} , so a ladder in β_{sub0} separates the hypotheses without touching the solver. Four runs, identical but for `-beta_sub0`, each 3406 steps and ~ 2 control-point spacings of meniscus travel. Batch 2026-08-25__18.14.15_beta_ladder, all four completed:

$\times$	<code>-beta_sub0</code>	β_{bare}	predicted ratio	measured inner	measured outer	$\beta_{\text{fit}}/\beta_{\text{bare}}$
0.10	5.9216e4	1.8741e5	0.3160	0.3239	0.3082	1.0004
0.25	1.4804e5	2.7623e5	0.5359	0.5444	0.5254	1.0015
1.00	5.9216e5	7.2036e5	0.8220	0.8253	0.8126	1.0017
4.00	2.3686e6	2.4968e6	0.9487	0.9471	0.9429	1.0000

The prediction is confirmed. The ratio moves $0.32 \rightarrow 0.95$ — a $2.9\times$ swing — and tracks the predicted curve to 2.3% worst case. The two rival hypotheses are dead: a flat 1.00 (the corrections operating as intended) and a flat 0.82 (a constant factor from elsewhere) are both excluded by every arm.

Fitting β with $d_{0,\text{sub0}}$ held fixed — the ladder has only one vapor BC per arm, so the two-parameter fit is ill-conditioned there, and d_0 was already measured to 0.06% on the nine-run 3×3 batch and does not depend on β — gives

$$\frac{\beta_{\text{fit}}}{\beta_{\text{bare}}} = 1.0009 \pm 0.0007 \quad \text{over a } 40 \times \text{ range in } \beta_{\text{sub0}}. \quad (12)$$

$C_{\text{corr}} = 0$ to better than a tenth of a percent. The correction this implies is §7.

7. What the correction should be

7.1 The exact statement

The realized coefficient obeys, exactly and with no further asymptotics required,

$$\beta_{\text{realized}} = \beta_{\text{sub0}} + \Delta, \quad \Delta \text{ from Eq. (6)}, \quad (13)$$

verified as $\beta_{\text{fit}}/\beta_{\text{bare}} = 1.0009 \pm 0.0007$ over a $40\times$ range in β_{sub0} . This is stronger than a scaling argument: not an estimate of an unknown correction with a fitted $O(1)$ prefactor, but a closed form for a known spurious term, read out of the source and confirmed to a tenth of a percent.

7.2 Why a fixed multiplicative factor cannot work

A tempting shortcut is to absorb the observed 0.822 into β_{sub0} . It fails, because 0.822 is not a constant — it is $\beta_{\text{sub0}}/(\beta_{\text{sub0}} + \Delta)$, and Δ does not scale with β_{sub0} . Calibrating the factor at the $1.00\times$ arm (pass $0.7835\beta_{\text{target}}$) and carrying it to the others:

arm	$\Delta/\beta_{\text{target}}$	realized/target	error
0.10 $\times$	2.165	2.948	+195%
0.25 $\times$	0.866	1.649	+65%
1.00 $\times$	0.217	1.000	0%
4.00 $\times$	0.054	0.838	−16%

Carried to the Molaro configuration it is -4.4% on the $D_v \times 10$ arm. The *sense* of such a factor is ambiguous too: $\Delta/\beta_{\text{target}} = 0.7835$ and $\beta_{\text{sub0}}/\beta_{\text{bare}} = 0.822$ are different numbers, and since $\beta_{\text{realized}} > \beta_{\text{sub0}}$ the model is too *resistive*, so any compensating factor is below one, not above it.

And in an important regime no factor exists at all. At Molaro’s $\epsilon = 1.2 \times 10^{-7}$ m with $\alpha_c \rightarrow 1$, $\beta_{\text{target}} = 7.94 \times 10^3$ while $\Delta = 1.79 \times 10^4$: Δ exceeds the target by $2.3\times$, so pre-compensation would need a *negative* `-beta_sub0`, which the option parser rejects. That is exactly the attachment-limited end wanted for a Demmenie replication.

7.3 The correction, and why it is a one-line change

Two operations are algebraically identical:

- pass $\beta_{\text{sub0}} = \beta_{\text{target}} - \Delta$, or
- **remove the two a_2 terms from τ_{sub} and pass β_{target} .**

Both give $\beta_{\text{realized}} = \beta_{\text{target}}$. Removal is preferable on three counts: it happens once in the code rather than in every user’s every run; it cannot be silently invalidated when someone changes ϵ or T ; and it still works when $\Delta > \beta_{\text{target}}$, where pre-compensation has no solution.

So the “correction factor” is not a number. It is the removal of a counter-term for a spurious effect this model does not produce — §5 is the reason, §6 the receipt.

8. Implementation

Authored in `enceladus_DSM`, back-ported to `lunar_regolith_DSM`. Enceladus is ahead: it already uses the T -corrected D_v in the a_2 term where lunar still uses the base 0°C value, and its `assembly.c` documents the $W_{\text{KP}} = \sqrt{2}\epsilon$ convention.

1. **Make the correction explicit and switchable** in both the scalar path (`enceladus_main.c:852`, `lunar_main.c:829`) and the pointwise one (`SubKinetics`), which carry the identical bracket:

```
PetscReal c_vap = a2 * eps / dv_T;
PetscReal c_therm = a2 * eps * drhovs_dT * lat_sub / k_eff;
if (!user.thin_iface_corr) { c_vap = 0.0; c_therm = 0.0; }
tau_sub = eps * lambda_sub * (beta_sub / a1 + c_therm + c_vap);
```

2. **Default `-thin_iface_corr 0`**, with the conditions for re-enabling it documented at the flag: only if the vapor diffusion is made two-sided, or an anti-trapping current is added, or a regime appears where $\epsilon v_n/D_v$ is not negligible. A physics switch, not a tuning knob.
3. **Fix the thermal coefficient** in the “on” branch rather than restoring it as-is. `drhovs_dT` is already computed and discarded at `lunar_main.c:735` ($8.1521 \times 10^{-5} \text{ kg m}^{-3} \text{K}^{-1}$ at -20°C); `k_eff` comes from `ThermalCond` at $\phi = 1/2$.
4. **Fix lunar’s base- D_v** in that term to match `enceladus`.
5. **Print $\beta_{\text{bare}}/\beta_{\text{sub0}}$ and the three bracket terms** in the run header beside τ_{sub} , so every future `outp.txt` states how far realized kinetics sit from requested.
6. **Validate:** re-run the $0.10\times$ ladder arm — 5.9 days simulated, 3406 steps — which predicts the measured ratio jumps $0.32 \rightarrow 1.00$.
7. **Recalibrate what depends on β :** `comp_eps.py` sizes ϵ from `beta_uns` (`eps_ceiling`, `kin_over_curv`); with the fix that relation stops being implicit. Re-derive ϵ for the standard -20°C geometries afterwards and confirm the mesh element counts do not move.

8. **Audit past conclusions** for absolute-rate dependence. Exponents and dimensionless ratios are unaffected; anything quoting a time-to-event or an absolute growth rate is off by the factor in §4 for its own (ϵ, T) .

9. Open

The spurious surface-diffusion term is the one $O(\epsilon)$ effect this note does not bound (§5). Settling it means either the one-sided asymptotics for this model — mapping onto Echebarria, Folch, Karma & Plapp, PRE **70**, 061604 (2004), whose $q(\phi)$ weighting is our ϕ_a — or adding the anti-trapping current of Karma, PRL **87**, 115701 (2001), which cancels it by construction:

$$\vec{j}_{\text{at}} = a_t \epsilon \rho_{\text{ice}} (\partial_t \phi) \frac{\nabla \phi}{|\nabla \phi|}, \quad (14)$$

added to the vapor flux, with a_t re-derived for M&F’s well exactly as a_1 and a_2 were. Neither is urgent: the effects anti-trapping cancels scale as $\epsilon v_n / D_v$, which is 2.7×10^{-14} here. It is worth doing before any claim that rests on distinguishing numerical from physical surface transport.

Priority is not the kinetics. For the Molaro campaign, removing attachment resistance *entirely* buys $1.28\times$ integrated over the neck range, against a $\sim 2\times$ gap — so no change to β closes it. The imposed far-field undersaturation is the larger lever: in arm B, $\sigma_\infty = -2.03 \times 10^{-3}$ against the neck’s own curvature driving force of $+6.57 \times 10^{-4}$, $3.1\times$ larger and opposed, so the neck surface is net-sublimating. The A/B and dom2/dom3 comparisons already isolate that.